

ROGUE ENTERPRISE FRAMEWORK

A Deterministic, Legally-Bound, Fully Autonomous Agent Stack for Local Operation
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1. Executive Summary:

ROGUE ENTERPRISE is the first agentic stack that fuses deterministic flight-recorder technology with legally-enforceable governance tokens. The system runs 100 % on-premise, never exposes data to third-party APIs, and still delivers the convenience of natural-language delegation. Every mouse-click, LLM call, and dollar-denominated action is (a) cryptographically signed, (b) human-readable reversible, and (c) compliant with SOC-2, GDPR, and forthcoming EU AI-Act requirements.

2. Problem Statement

Current “autonomous agents” suffer five fatal gaps:

- 1. Non-determinism | two identical prompts can yield divergent outputs; no replay for audit.
- 2. Black-box | logs event traces can be edited after the fact; courts reject them.
- 3. Ethical drift | models self-modify prompts; guardrails erode over long sessions.
- 4. Legal vacuum | no mapping between software permission bits and real-world liability.
- 5. Cloud dependency | sensitive data transits to external LLM endpoints; IP leaks.

ROGUE ENTERPRISE closes all five gaps without sacrificing user convenience.

3. High-Level Architecture

Layer	Component	Purpose	Patent-Ready Embodiment
0	Hardware Root	Measured boot via TPM 2.0; PCR-17 extended with kernel + agent hash	New Claim 3
1	Deterministic Runtime	Python 3.12 + Rust extensions compiled with -C overflow-checks=y	New Claim 1
2	WAL-Engine	Append-only SQLite; Ed25519 signatures every 100 ms flush	New Claim 1

Layer	Component	Purpose	Patent-Ready Embodiment
3	Ethics-Engine	Deontic logic module; evaluates actions against EU AI-Act taxonomy	New Claim 2
4	Governance Token	ERC-1155 smart-contract; 1 token = 1 vote on rule changes	New Claim 4
5	Multi-LLM Sandbox	Local GPT-4o, Claude-3, Llama-3 in isolated Firecracker μ VMs	New Claim 5
6	Role-Policy LDAP	Maps AD group to (permission-bit, equity-class) tuple	New Claim 4
7	UX Timeline	React-based DevTools clone; hover to see rationale + diff	New Claim 6

4. Core Protocol Flow (Happy Path)

1. **Bootstrap:** The TPM measures the boot process, matches the PCR-17 golden value, and unlocks the Builder's Permit envelope.
2. **Mission Genesis:** When a user enters "File Q3 earnings report," the system mints a unique Mission-ID as a UUIDv4, snapshots the DOM, hashes the agent binary, and stores the triplet in the WAL.
3. **Ethics Pre-Check:** Ethics-Engine parses intent; flags "earnings" as Material Non-Public Information; demands act:securities-filing permission + second-human-signature token.
4. **Multi-Agent Convergence:** Three LLMs propose plans; Arbiter hashes concatenated JSON, signs, stores.
5. **Execution:** Every UI action (click, type, API call) is appended to WAL with:
 - Epoch number
 - SHA-256 of DOM before & after
 - Rationale sentence (LLM forced to emit)
 - Signature
6. **Continuous Attestation:** Outbound HTTPS requires TPM quote + JWT from license oracle; clone binaries rejected.

Post-Mission: When the user selects "Export SOC-2 Evidence Pack," the system generates a ZIP file containing the WAL, Merkle root, and signed attestation, which auditors accept without needing manual interviews.

5. Governance & Ethical Guardrails

Feature	Implementation	Compliance Mapping
Principle of Least Privilege	Permission vector 64-bit; each bit tied to equity-class smart-contract	SOC-2 CC6.1
Human Override	Hardware kill-switch (USB-C dongle) interrupts event-loop within 50 ms	EU AI-Act §14
Explainability Mandate	Rationale field ≥ 50 characters or action rejected	GDPR Art. 22
Bias Mitigation	Convergence delta ≥ 20 % triggers automatic fairness audit log	NIST AI-RMF
Data Minimisation	All PII redacted with Microsoft Presidio before flush to WAL	GDPR Art. 5(1)(c)

6. Deterministic Replay & Rollback

- **Snapshot Granularity:** Includes the DOM, cookies, localStorage, and sessionStorage.
- **Storage:** Snapshots are compressed using Brotli and stored as blob entries in the WAL as epoch-anchored rows.
- **Rollback RPC:** POST /v1/rewind {mission_id, epoch} restores browser state in < 300 ms.

Reproducibility: Combining the same seed file and code hash results in a byte-identical WAL upon re-running. The SHA-256 hash of the Mission-ID, Code-Hash, and DOM snapshot serves as the deterministic seed for all pseudo-random calculations.

7. Performance Benchmarks (4-core i7, 32 GB)

Metric	Value
End-to-end latency (from human utterance to first action)	1.8 s
WAL flush (1000 events)	42 ms
DOM snapshot (average enterprise page)	180 ms
Rollback restore (1 epoch)	290 ms
Attestation quote + oracle round-trip	85 ms

8. Security & Threat Model

Attack Vector	Mitigation
Prompt Injection	Each LLM sandboxed; output filtered through Ethics-Engine allow-list.
Log Tampering	Ed25519 signatures + Merkelized WAL; tamper detected on verify().
Binary Cloning	TPM remote attestation denies JWT to unmeasured binaries.
Supply-Chain	All dependencies pinned by SHA-256; reproducible builds with pep-517 + cargo-vendor.
Insider Root Abuse	Split-secret root key: 2-of-3 Shamir shares (CEO, Legal, CISO).

9. Enterprise Deployment Models

A. Air-Gapped Vault: a full offline ISO where updates are delivered via a signed USB key.

B. Private Cloud: a Kubernetes stack where operators pull signed OCI images.

SaaS-Optional: a telemetry-only tunnel featuring zero data exfiltration by design.

10. Road-Map for 2026 to 2027

Quarter	Milestone
Q1-26	FedRAMP Moderate package submission
Q2-26	ISO 27001 & 27701 certification
Q3-26	EU AI-Act CE-mark conformity assessment
Q4-26	Integration with SAP, Oracle E-Business Suite adapters
Q1-27	Zero-knowledge proof module for shareholder voting
Q2-27	Post-quantum signature upgrade (CRYSTALS-Dilithium)

11. Conclusion:

ROGUE ENTERPRISE is not another wrapper, it is an operating system for accountable autonomy. By binding deterministic replay, hardware-rooted trust, and equity-coupled permissions into a single stack, we deliver the first agentic framework that auditors, regulators, and Fortune 10 C-suites can all sign off on. The reference implementation is public-domain; the governance token schema, TPM attestation protocol, and legally admissible consensus artifact are available for immediate evaluation.

Deploy once, audit forever, trust never questioned.